Jordi Martínez-Muñoz Research Statement
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We often describe economics as the allocation of limited resources to unlimited needs — I would add, most of the time under uncertainty. Information is key to such decisions, and how people use it — or fail to use it — is crucial to understanding behavior and designing better policy. My research, at the intersection of experimental and behavioral economics, studies how people use information when making decisions and forming beliefs under uncertainty.

When exploration/exploitation decisions do not track changes in information. Many sequential decisions, such as research projects, investments and partnerships, involve choosing among alternatives whose quality is learned only through experience. Standard models of sequential experimentation predict that people switch between alternatives only when they learn something new. In my job market paper, "Beyond the Exploration–Exploitation Trade-off: The Behavioral Structure of Switching between Alternatives," I test this prediction in an incentivized experiment with 300 subjects, based on the canonical one-armed bandit problem (Robbins, 1952; Gittins and Jones, 1974): repeated choice between a familiar arm, with a known payoff distribution, and an unknown arm. A reward from the unknown arm reveals that it is better, while outcomes of the familiar arm reveal nothing, so the model predicts single switching: try the unknown arm until it pays or enough failures accumulate, then stay with one arm for good.

I find that about half of subjects over-switch: they return to the unknown arm after switching away from it, even though no new information can justify the return. Over-switching persists after uncertainty is resolved: once the unknown arm pays and its quality is known, 39% of subjects still switch away from it at least once. Part of this switching is a reaction to outcomes that carry no information: among over-switchers, a second reward in a run on the familiar arm lowers the probability of switching away by about 10 percentage points. These deviations distort exploration and are costly: over-switchers explore bad unknown arms about 40% more, find good ones later, and earn 20 to 25% less than single-switchers.

The data reject random switching, win-stay/lose-shift, reinforcement learning and treating the familiar arm as of unknown quality, and rule out utility curvature as the leading explanation of over-switching. I then test whether over-switching reflects a single-switching plan executed with noise that comes from reacting to round-by-round outcomes, or a valuation of the arms that differs from the model's (an incorrect model of the problem or

non-standard preferences). To do so, I run two treatments that let subjects choose plans that limit their reaction to realized outcomes: choosing an arm together with the number of rounds to draw from it, and choosing an automatic rule before observing any outcomes.

In the subsample that faced these treatments, about half of the over-switchers choose single switching in each, consistent with noisy execution; the rest keep switching, and about a quarter do so under both, consistent with a valuation that differs from the model's. The subjects who respond the same way to both treatments already differ in the baseline free-switching treatment: those who single-switch under both switch less often and react less to failures than those who keep switching under both. I use these differences to classify all over-switchers in the baseline treatment and find that a single-switching plan that deviates on at most one decision in ten best predicts the choices of 61% of those relatively closer to the group that single-switches under both treatments, but only 25% of those relatively closer to the group that keeps switching.

These results motivate two ideas that are currently work in progress. The first one is part of **extensions to strategic experimentation**, in which two subjects face a common unknown arm — its true quality is the same for both — and observe each other's choices and outcomes; I test whether the familiar arm's payoff variability (deterministic vs. stochastic) shapes the free-riding under-exploration predicted in strategic experimentation (Keller, Rady, and Cripps, 2005). The second project is part of **extensions to biased beliefs about random sequences.** I test whether gambler's-fallacy and hot-hand-type distortions in forecasting a known random process — well documented in belief elicitation (Benjamin, 2019) — carry over to a consequential, incentivized choice between lotteries, or whether such distortions are confined to forecasting within a single lottery. Together, this agenda asks when and why people use information—or respond to inputs that are not informative—when making decisions and forming beliefs under uncertainty.

Does Law Have an Informative Function? Information can also come from institutions: when the health risks of an activity are uncertain, people may use its legal status as a signal of those risks (the informative function of the law; Dharmapala and McAdams, 2003). In joint work with Fernando Briseño, José Antonio Márquez, and Víctor Pérez, "Legal Information as an Input to Health Risk Assessments: Evidence from E-Cigarettes in Latin America," we test this in the domain of e-cigarettes, with a randomized information experiment in Mexico, Argentina, and Chile, three countries with different and recently defined regulatory regimes. Randomly informing respondents who report being unaware of their country's current regulation about its legal status identifies the causal effect of legal

information on perceived health risk.

With 3,600 respondents, we find that learning the legal status leaves perceived health risk largely unchanged across measures (relative risk, relative addiction, and absolute risk relative to cigarettes), including among respondents whose prior beliefs are contradicted by the information. Yet beliefs across countries align with their regulatory regimes: perceptions are more negative in countries where e-cigarettes are banned, even among respondents who report being unfamiliar with the regulation or hold incorrect beliefs about it.

Looking ahead, I plan to study how exploration changes when an agent's own ability affects outcomes, or when each arm's payoff depends on an observable context that varies from round to round (contextual bandits). Building on the legal-information project, I also plan to study how people update beliefs on contested topics when given information from authoritative sources.

References

Benjamin, D. J. (2019). Errors in Probabilistic Reasoning and Judgment Biases. In B. D. Bernheim, S. DellaVigna, & D. Laibson (Eds.), *Handbook of Behavioral Economics: Applications and Foundations*, Vol. 2 (pp. 69–186). Elsevier.

Dharmapala, D., & McAdams, R. H. (2003). The Condorcet Jury Theorem and the expressive function of law: A theory of informative law. *American Law and Economics Review*, 5(1), 1–31.

Gittins, J. C., & Jones, D. M. (1974). A Dynamic Allocation Index for the Sequential Design of Experiments. In J. Gani (Ed.), *Progress in Statistics* (pp. 241–266). North-Holland.

Keller, G., Rady, S., & Cripps, M. (2005). Strategic Experimentation with Exponential Bandits. *Econometrica*, 73(1), 39–68.

Robbins, H. (1952). Some Aspects of the Sequential Design of Experiments. *Bulletin of the American Mathematical Society*, 58(5), 527–535.